Table 3. Screen and supporting analyses

A. BBJ discoveries and FinnGen GD comparison

BBJ discovery gene	BBJ OR (95% CI)	BBJ *P*	FinnGen GD OR (95% CI)	FinnGen GD *P*	Focused support	GD PP.H4
HLA-A	1.98 (1.73–2.27)	2.60×10^{-23}	1.08 (0.77–1.53)	0.648	No	<0.001
HLA-DQA2	2.39 (1.95–2.93)	8.30×10^{-17}	1.90 (1.64–2.20)	1.41×10^{-17}	Yes	<0.001
CTLA4	0.18 (0.11–0.27)	5.45×10^{-15}	0.17 (0.08–0.36)	2.15×10^{-6}	Yes	0.254
TSHR	0.12 (0.07–0.21)	1.09×10^{-14}	0.08 (0.05–0.12)	3.19×10^{-32}	Yes	0.264
C4A	0.45 (0.35–0.57)	2.23×10^{-10}	0.41 (0.26–0.63)	4.97×10^{-5}	Yes	<0.001
HSD3B7	0.29 (0.18–0.47)	2.04×10^{-7}	0.90 (0.72–1.13)	0.372	No	0.011
TUBB	1.49 (1.28–1.74)	3.14×10^{-7}	3.28 (2.09–5.17)	2.68×10^{-7}	Yes	<0.001
VKORC1	0.13 (0.06–0.29)	6.35×10^{-7}	NA	NA	NA	0.011
TNFSF14	0.63 (0.52–0.76)	1.48×10^{-6}	0.77 (0.65–0.92)	0.00361	Yes	0.303
PRSS36	6.00 (2.87–12.57)	2.02×10^{-6}	1.40 (1.00–1.96)	0.0490	No	0.063
MAPKAPK5	0.977 (0.967–0.987)	5.28×10^{-6}	0.990 (0.929–1.055)	0.751	No	0.041
PSMB8	1.25 (1.13–1.38)	6.86×10^{-6}	0.96 (0.85–1.08)	0.512	No	<0.001
IFNGR1	2.10 (1.51–2.91)	9.41×10^{-6}	1.62 (1.18–2.24)	0.00330	Yes	0.313

B. Additional FinnGen GD discoveries

Additional gene	GD IVs	GD OR (95% CI)	GD *P*	BBJ PP.H4	GD PP.H4	GO strict PP.H4
APOM	1	0.05 (0.04–0.06)	1.43×10^{-108}	<0.001	<0.001	0.227
NEU1	1	0.04 (0.03–0.06)	1.43×10^{-108}	<0.001	<0.001	0.061
IGLV2-18	1	0.56 (0.48–0.66)	7.52×10^{-12}	0.029	0.020	0.100
IGLV3-12	1	1.20 (1.14–1.26)	7.52×10^{-12}	0.019	0.022	0.101
TNF	1	0.60 (0.51–0.71)	4.65×10^{-10}	<0.001	<0.001	<0.001
HIST1H4E	1	2.54 (1.88–3.43)	1.51×10^{-9}	<0.001	<0.001	0.007
LTA	1	0.46 (0.35–0.61)	3.75×10^{-8}	<0.001	<0.001	<0.001
LTB	1	0.18 (0.10–0.36)	4.89×10^{-7}	<0.001	<0.001	<0.001
C4B	2	1.63 (1.31–2.03)	1.10×10^{-5}	<0.001	<0.001	0.311

C. Frequency-sensitive results

Gene / GD analysis	Reference OR (95% CI)	Reference *P*	Cohort-frequency OR (95% CI)	Cohort *P*	Interpretation
C4B	1.63 (1.31–2.03)	1.10×10^{-5}	1.75 (1.31–2.33)	1.48×10^{-4}	Heterogeneous; loses screen significance
HLA-G	0.86 (0.72–1.03)	0.107	0.60 (0.47–0.75)	1.72×10^{-5}	Weight-sensitive; single-variant dependence

Panel A: focused support requires concordant direction and $P < 0.05/13$; 7 of 13 qualify. NA is unavailable, not a negative result. Panel B: nine additional reference-frequency FinnGen GD discoveries at $P < 0.05/2{,}544$; post hoc and not independent of the full screen. Seven of the nine are instrumented within the extended MHC region on chromosome 6. *APOM*/*NEU1* share rs3101018; *IGLV2-18*/*IGLV3-12* share rs5759410. The lead *IGLV* expression variant is absent in BBJ, weakening interpretation of its low PP.H4. Panel C: *HLA-G* was added because of frequency sensitivity, not primary discovery. Neither *C4B* nor *HLA-G* gains strong GD colocalization. All 24 genes and five datasets are available in the workbook.